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Size matters – effect of screen setup on muscle activity and posture in computer work

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ABSTRACT

This research aims to investigate and compare how computers work on five different workstation setups of multiple or large screens and viewing distances (27-inch and 49-inch at 70 or 90 cm viewing distance) affect muscular load in neck/shoulders, working postures, and perceived exertions of non-presbyopic participants. The electrical activity in the trapezius shoulder and trapezius neck muscles, and work postures of the head and back were recorded with objective, technical methods. The perceived exertions in the neck/shoulder and eyes, and the participants' opinions of the different workstations, were collected. Overall, the 49-inch curved screen with a viewing distance of 90 cm was quantitatively recorded and qualitatively rated as the best setup of all five workstations included in the study. The 49-inch screen was also objectively and subjectively rated to be better than 2×27-inch screens. The 2×27-inch V-formation setup was subjectively favoured compared to the 2×27-inch L-formation setup.

Practitioner summary: Practitioners at the Occupational Health Services ask for counsel on how to place large or multiple computer screens. By using quantitative and qualitative methods, we propose recommendations on size, viewing distance, and screen setup for non-presbyopic humans to reduce the risk of visual and physical ergonomic complaints during computer work.

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KEYWORDS
Computer vision syndrome; musculoskeletal disorders; neck and shoulder; visual ergonomics

Introduction

Computers have become a natural part of many people's everyday life, both at work and leisure time. Multiple monitor configurations or large computer screens have become increasingly common, for the promotion of productivity by providing more screen space (Gallagher et al. 2021). This means altered demands on the visibility of the work object (text size and contrast), adequate screen height, ergonomically advantageous working positions, and corrective glasses (Woo et al. 2016; Anshel 2006). In addition, glare-free lighting is also required to provide sufficient illumination (Boyce 2003).

Screen and display work is visually demanding work, i.e. when the accommodation is high in near work during small viewing angle and extended time (Rosenfield and Portello 2016; Coles-Brennan et al. 2019). Similarly, when viewing a flat screen during accommodation and convergence, the three-dimensional (3D) visual impressions of our central visual field will be reduced, thus causing more eye strain (Glimne et al. 2015). Furthermore, it is easier for the eyes to converge downwards than looking straight ahead (Jaschinski et al. 1998). Increased visual demands can cause a variety of symptoms referred to as eye problems, Computer Vision Syndrome (CVS), or digital eyestrain (Thorud et al. 2012; Kaur et al. 2022). CVS is a combination of eye and vision problems associated with the use of computers, e.g. eye fatigue, red eyes, burning eyes, watery eyes, headache and neck/shoulder pain (Thorud et al. 2012; AOA XXXX).

Several studies have reported a high prevalence of work-related musculoskeletal disorders (WMSDs) in the neck/shoulder and wrist regions among computer workers (Demissie et al. 2024; Szeto et al. 2005). Neck and upper extremity pain has a multifactorial origin including both organisational and psychosocial factors (e.g. high work demands and time pressure) as well as physical load and individual factors (e.g. gender, BMI and previous illness history) (Eijckelhof et al. 2013; Wahlstrom 2005; Vinothini et al. 2018). The physical

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load in this context is mainly static postures, which generate a sustained low-level muscle activity with small opportunities for muscle recovery. This imposes biomechanical strain on ligaments, joint capsules, muscle fibres and other musculoskeletal structures (Szeto et al. 2005). Thus, symptoms of WMSDs among computer users may be an interaction of physical, psychosocial and individual risk factors (Vinothini et al. 2018).

Large or multiple screens may involve awkward head positions when looking at the part of the screen farthest to the side, or a screen placed far out laterally. Moreover, the head rotation during computer work when using two screens increases the load on the neck and shoulders (Shamsul Bahri et al. 2016; Szeto et al. 2014). That there is an association between visual discomfort, eyestrain and muscle activity in the neck and shoulders has also been demonstrated (Zetterberg et al. 2017; Mork et al. 2020; Seghers et al. 2003). Furthermore, a poorer visual ergonomic conditions in workplace environments can lead to increased visual fatigue and reduced work efficiency due to stress on the visual function (Anshel 2006).

There are some recommendations to decrease ergonomic problems during computer work. They are based on studies carried out using smaller screens (below 20 inches) in the early 2000s, on how to place the screen in relation to the user:

- The height of a screen should provide a viewing angle from the horizontal plane in the primary position[1] (Wright 2003) to the centre of the screen that is 15–30 degrees downwards (Fostervold et al. 2006; Burgess-Limerick et al. 2000; Allie et al. 2005). This usually corresponds to having the top edge of the screen aligned with the eyes vertically when looking horizontally in primary position.
- The eyes upward gaze angle should not exceed 15° from the horizontal plane in primary positions because there is a demonstrable negative impact on the co-operation between the eyes that causes an extended neck (Govender et al. 2009).
- A viewing distance to the centre of the screen that is 60–70 cm is preferable to a viewing distance of 90–100 cm, as the latter distance has been reported to be more visually demanding (Jaschinski et al. 1999).

Previous work has demonstrated that using flat or curved 34-inch screens compared to smaller ones (mainly 22-inch screens) reduced neck problems, likely because more neutral neck postures were possible (Bartha et al. 2020). The same study also showed that people like to place curved screens slightly further away than flat ones, even if the screens were the same size. However, a larger curved screen can present other challenges, e.g. requiring more space and a longer viewing distance, which may be challenging to achieve with today's standardised table depth of 80 cm.

Although there have been many studies on visually demanding work in computer environments, there are only a few on how the use of large and multiple screens affects visual function, muscle activity and working posture. For example, it has been concluded that there are statistically significant differences in trapezius muscle activity and neck rotation between single-, L- and V-screen formations (Fang et al. 2023; Pokhrel and Hwang 2024). Likewise, in a review carried out by Gallagher et al. moderate evidence for increased neck rotation was found when working with multiple screens (Gallagher et al. 2021). In the same review article, though, the authors could not present sufficient evidence to guide current practices either for eye strain or neck muscle activity. Recently, large and curved screens as large as 49 inches have become available on the market. These screens form a semicircle in front of the worker. The effects of curved displays on legibility and visual fatigue were investigated by Park et al. They concluded that moderately large, curved screens were more legible and less visually fatiguing than a flat display (Nordander et al. 2004). Still, there is limited knowledge on how large and curved screens affect the working posture and muscle activity in the neck-/shoulder region compared to more standard screen sizes.

The aim of the current study was to investigate and compare how computers work on large and multiple screens at different viewing distances, including high/low contrasts, and affect:

The working postures, muscle activity in neck/shoulders, perceived exertions, and subjective ratings of non-presbyopic participants.
Based on our findings and previous literature, we propose recommendations to reduce the risk of visual and physical ergonomic complaints during computer work.

Materials and methods

Study design and participants

Twenty-nine non-presbyopic students, 15 women and 14 men, volunteered to participate in the study. Their

median age was 24 years (range 19–28), height 176 cm (range 160–198), and weight 75 kg (range 47–95). The inclusion criteria were an age of 18–35 years and an approved visual test (see the next section 'Visual test for inclusion in the study'). The study took place in a laboratory setting where the participants performed two computer work tasks at five different workstations. Each station contained a separate set consisting of a computer and a screen/screens. During the work tasks, the physical workload in terms of the head and back postures, along with the electrical activity of the upper trapezius muscles, was recorded with objective, technical methods. The order of the workstations was randomised.

The Swedish Ethical Review Authority approved the study (No. 2023-01450-01). All participants gave their written informed consent to participate in the study.

Visual test for inclusion in the study

Each student underwent a visual test for inclusion in the study. Those with subjective symptoms that could be related to accommodation and vergence disorders (i.e. visual and ocular discomfort when performing near visual tasks) were excluded. The participants also had to be free of ocular pathology and not be under treatment with a medication known to have an effect on visual acuity and/or binocular vision. The participants also had to achieve a corrected visual acuity of 1.0 (i.e. 6/6 Snellen visual acuity), both monocularly and binocularly. Finally, only participants with good stereo acuity (TNO test, random-dot stereographs, of 60 s or better) and normal colour vision (approved Ishihara test) were included (red-green colour vision test, Ishihara test).

Workstations and design

The five separate workstation setups were as follows (Figure 1):

- One 27-inch screen (denoted 27), with a viewing distance of 70 cm to the centre of the screen.



Figure 1. The five setups of the workstations:
27 = one 27-inch screen with a viewing distance of 70 cm.
27L = two 27-inch screens, with one screen in front of the participant and the other to the left. The angle between the screens was 140° and the viewing distance was 70 cm.
27V = two 27-inch screens, with the screens placed in a V in front of the participant. The angle between the screens was 140° and the viewing distance was 70 cm.
49–70 = one 49-inch screen with a viewing distance of 70 cm.
49–90 = one 49-inch screen with a viewing distance of 90 cm.

- Two 27-inch screens (denoted 27L), with one screen in front of the participant and the other one to the left. The viewing distance was 70 cm.
- Two 27-inch screens (denoted 27V), with the screens placed in a 'V'. The viewing distance was 70 cm.
- One 49-inch screen with a viewing distance of 70 cm (denoted 49–70).
- One 49-inch screen with a viewing distance of 90 cm (denoted 49–90).

The computers ran Windows 10 with at least 4 GB of RAM and a generation 3 Intel processor. The keyboard model was Dell KB216 – QWERTY, the mouse model was Dell MS-116, and the screen models were 27 Dell SE2723DS IPS DP/2HDMI HAS 1440p and 49 Samsung S49A950 Curved 2HDMI HAS High 1440p. The visual angle was 46.1×24.2 degrees for the 27-inch screens, 81.2×24.2 degrees for the 49-inch screen with the viewing distance of 70 cm, and 67.4×18.9 degrees for the 49-inch screen with the viewing distance of 90 cm. The screen resolution was 2560×1440 pixels for the 27-inch screen and 3840×1080 pixels for the 49-inch screen.

An office chair without armrests was adjusted according to the height and comfort of the participant. The same chair with the same individual adjustments was used for all five workstations to ensure that the seating conditions were similar for all workstations.

The desk height was adjusted to fit the participant, and the same height was used for all workstations.

The viewing angle from the horizontal plane in primary position to the centre of the screen was adjusted for each workstation using a spirit level. The group mean of the viewing angle was 20° for 27, 27 L and 27 V and was 21° for 49–70, with a common range for the four workstations that was 16° to 26°. For workstation 49–90, the group mean was 17° with a range between 13° and 20°. The group mean of the height of the screen (from the desk surface to the upper edge of the screen) was 46 cm with a range between 40 and 50 cm for all five workstations. The differences in the group means are due to individual factors such as differences in upper body length and arm length.

The lighting conditions in the room provided luminances between 30 and 125 cd/m^2 on and behind the table area (including the screens), except for the mouse and keyboard, which provided luminances of about 5 cd/m^2. The average illuminance levels for all five workstations' desks were 550 lux. These conditions were similar for all five workstations (Figure 2).

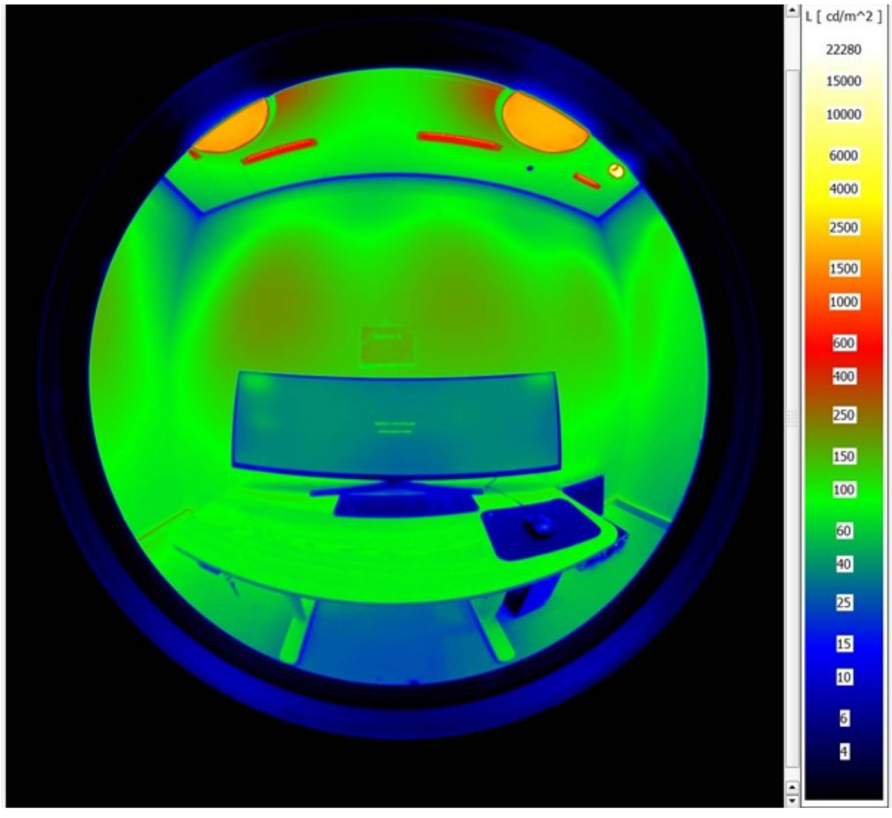


Figure 2. The luminance setting for all five workstations. The ceiling luminaires were not in the visual field during the tests.

Computer work tasks

Participants were instructed to conduct two tasks at each workstation, described in detail below. In the first *visual search*, participants were asked to find a grating that had a slightly different orientation than the other gratings in a search array. In the second, *drag and align*, involved finding and moving a grating to a specific position and orientation using the mouse and keyboard. The work tasks were implemented in Python using PsychoPy (Standalone v. XX). Code to run the tasks is publicly available on GitHub: https://github.com/marcus-nystrom/ScreenSizeMatters.

'Visual search'

Stimuli

Each search array consisted of 25 gratings distributed within a 5 × 5-degree window on the computer screen for both the 27-inch screens and the 49-inch screens, presented on a mid-grey background (see Figure 3). The search array size in cm was 7.4 × 7.4 cm for the viewing distance of 70 cm and 8.9 × 8.9 cm for the viewing distance of 90 cm. Gratings were generated in PsychoPy through the GratingStim function and had a size of 1 degree and a spatial frequency of 5 cycles per degree. The contrast of the gratings was either high (contrast = 0.5) or low. In the latter case, contrast was controlled with a 1-up, 3-down staircase to keep the orientation of the gratings barely visible.

Procedure

Participants were instructed to identify the grating with an orientation deviating from all the other gratings in the search array, and then press the spacebar key. The gratings were then replaced by dots, and the participants were asked to left-click with the mouse on the dot located at the position where the deviating grating was presented earlier. Participants performed the search at five distinct locations on the screen (the centre and the corners). At each position, they conducted searches within a 60-s interval, whereafter the search moved on to a new, randomly drawn screen location. Participants first completed the search task with high contrast targets and then with low contrast targets. In the low contrast condition, the first search had a duration of 70 s instead of 60 s to allow participants to approach their person-specific threshold in the staircase. Even if participants ran out of time, they were allowed to complete the last search at each of the five screen locations. Thus, the total search time typically slightly exceeded the set durations (60 s or 70 s).

Finally, to establish a *baseline* where no visually demanding task or mouse movements were performed, the participants were asked prior to the visual search tasks asked to look somewhere within the search array (without an explicit instruction to search for something). The stimulus presentation was otherwise identical with the high contrast search, except that no dots were shown to prompt a mouse click. Consequently, there were in total 5 × 60 s (five positions of the search array for 60 s each) where participants looked at search arrays during the baseline condition, 5 × 60 s where participants searched for high contrast targets, and 5 × 60 s (+10 s for the first search) where participants searched for low contrast targets.

In total, this part of the study took about 17–19 minutes to complete for each participant at each workstation.



Figure 3. An example of the stimulus for the visual search task. Participants were asked to find the grating with the deviating orientation. In the example, it is found at row 2 and column 4.



Figure 4. Stimuli for the drag and align task. The participant was to align the position and orientation of the grating with the red circle. This was achieved by dragging the grating with the computer mouse and rotating it with the left and right arrow keys.

'Drag and align'

Stimuli

The stimulus consisted of a grating with the same properties as in the visual search task, and a circle with a radius of 0.4 degrees with a red outline presented on a mid-grey background (Figure 4). A line was drawn through the circle to give it an orientation. The positions of the grating and the circle on the screen(s) were randomly drawn from a uniform distribution of possible screen locations (excluding those where the grating and the dot would not be fully visible). The orientation (in degrees) of the line within the circle was randomly drawn from a uniform distribution, U (0, 180).

Procedure

The participant was instructed to click (with the left mouse key) and drag the grating onto the circle so that their centres overlapped. The participant was also asked to use the left and right arrow keys on the keyboard to rotate the grating so that its orientation aligned with the direction of the line. When the participants had completed the positioning and orienting of the grating, they could proceed to the next trial by clicking a button in the lower right corner of the screen. They had 5 minutes to complete the work task.

Together, the 'visual search' and 'drag and align' tasks lasted about 22–23 minutes. Between each workstation, participants had a small break for about 6–8 minutes, except after the third station, when they had a 30-minute break.

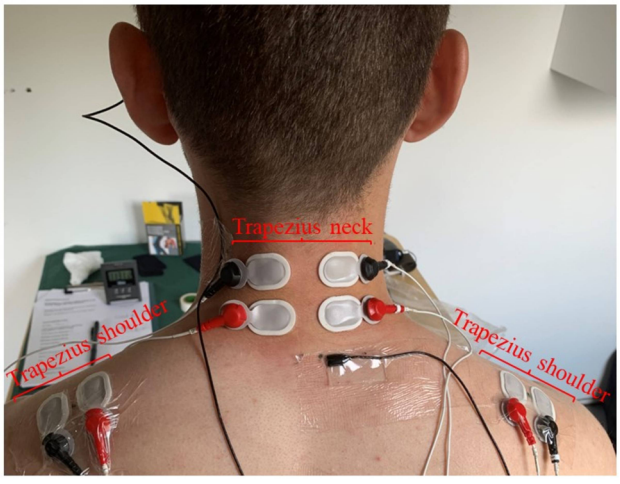


Figure 5. The four pairs of electrode positions during the recording of the electrical activity at two different locations of the right and left trapezius muscles.

Surface electromyography (EMG)

The participants' electrical activity was recorded bilaterally by four pairs of Ag/AgCl electrodes (Ambu Neuroline 720; Ambu A/S, Ballerup, Denmark) at two different locations of the trapezius muscles: neck and shoulders (Figure 5). The skin was cleaned with alcohol, and two pairs of electrodes were attached to the skin on the descending parts of the right and left upper trapezius, 2 cm lateral to the midpoint on the line between the seventh cervical vertebra and the lateral edge of the acromion (trapezius shoulders; Figure 5). The other two pairs were attached on both sides of the third and fourth cervical vertebrae (trapezius neck, Figure 5). The centre-to-centre distance of the electrodes was 22 mm (trapezius shoulders) and 28 mm (trapezius neck), respectively, with an active diameter of 6 mm.

Prior to the computer work tasks, the participants performed three maximal voluntary contractions (MVCs) for the two muscle locations. This consisted of arm abductions with one arm at a time raised to 90° in the scapular plane, to achieve the maximal voluntary electrical activity (MVE) level of the muscles studied. The participants stood in a stable position with their feet somewhat apart, holding the trunk straight. A strap anchored to a wooden plate was placed just proximal to the elbow, and the participant was instructed to pull as hard as possible without jerking (Hansson et al. 1997). An experienced instructor supervised the participants to maintain each contraction for about 5 s and encouraged them to perform at their best. After the recording, the signals were amplified, filtered (10–400 Hz) and sampled at a rate of 2048 Hz, and stored on a 2-GB memory card in a Mobi-8 data logger (TMS International, The Netherlands). After recording, the data were transferred to a computer for further processing, quality assurance, and analysis in the EMINGO software suite, developed by the Division of Occupational and Environmental Medicine in Lund, using MATLAB version 2019b (Math Works INC., USA). The EMG signals were digitally band-pass filtered (30–400 Hz) to remove electrical signals from the heart, and were notch filtered (mains frequency, 50 Hz, and all harmonics). The root mean square value was calculated for periods of 0.125 s, and the noise was subtracted in a power sense (Jensen et al. 1993). A moving window with a width of 0.5 s was used to find the highest EMG activity resulting from the three efforts of MVCs for each kind of contraction (Jensen et al. 1993; Hansson et al. 2003). The muscular activity during the tasks was normalised to the maximal voluntary electrical activity (MVE) recorded during the MVCs.

Inclinometry

Postures of the head and upper back were recorded using triaxial accelerometers (Logger Teknologi HB, Åkarp, Sweden) in combination with a data logger (Logger Teknologi HB, Åkarp, Sweden) with a sampling frequency of 20 Hz (Hansson et al. 2006; Gunnar 1990). The accelerometers were mounted with double-sided adhesive tape to the middle of the forehead just above the eyebrows, and to the upper back to the right of the spine at the level of C7. The reference posture for the head and back (0° inclination) was performed with the participant standing upright and looking straight ahead. All postures during the recordings were calculated in relation to the reference posture. After recording, the collected data were transferred to a computer for processing and analysis in the EMINGO software suit.

Perceived exertion

After carrying the two work tasks out at each workstation, the participants were asked to rate their eye complaints (e.g. eye fatigue), visual impairment (at the moment they experienced it as most difficult), and perceived exertion in the neck and shoulders according to the Borg CR-10 scale (Nordander et al. 2013). The search array was adjusted to have the same angular size in degrees across all workstations. Thus, we do not predict a difference in perceived exertion for the different screens due to angular size and screen resolution.

Interview

The participants were interviewed after their participation. They were asked for their opinion of which workstation they thought were the best and worst, and their reasons and comments for that. Notes were taken by the research leaders.

Data analysis

The recorded data from the 'visual search' task and 'drag and align' task was merged (to reflect ordinary office work) and analysed as one work task (referred to as 'computer work'). Muscular load and postures during the computer work were compared across workstations. The muscular load was estimated by computing the 50th and 90th percentiles of the amplitude distributions (expressed as %MVE) and the proportion of muscular rest (% of time with an MVE <0.5%) for each participant and each workstation. This resulted in three values for each workstation and participant: one for each of the two percentiles and one value for muscular rest. Postures were also estimated by computing the 50th and 90th percentiles of angular distributions of head and back for each participant and each workstation. Group means were then calculated, which are the same calculations that have been made on data from numerous of field studies, to describe physical workload (Balogh et al. 2019; Ostensvik et al. 2009). For the percentiles, a higher value was considered more strenuous, while a higher value for muscular rest was considered less strenuous. To investigate the influence of contrast on physical workload, muscular load and head and neck postures were compared during the visual search task when targets had high (lasting on average 24 min per participant) and low (lasting on average 30 min per participant) contrasts. The number of mouse clicks during high and low contrasts were registered for each workstation and then summarised for each participant. A group mean was then calculated. Moreover, group means of perceived exertion in eyes and neck/shoulders were calculated.

The interview answers were categorised by one of the co-authors. First, the individual answers for each participant were entered into a text file. Second, the answers were categorised according to workstation. Finally, the categorisations within each workstation were further categorised into positive or negative reviews (Table 1). The participants stated some general comments such as 'the chair was comfortable', 'gradually tiring to sit' and 'strenuous'. However, these were not informative to our research questions and therefore not included in the categorisation process.

Statistical analysis

The statistical analysis was carried out in IBM SPSS Statistics Version 28 (SPSS, Chicago, IL, USA). A p-value of <0.05 (two-tailed) was considered significant. An overall comparison between the different workstations was carried out using Friedman's test. If Friedman's test was significant, pairwise comparisons between the five different workstations were performed using Wilcoxon signed rank test regarding physical workload and perceived exertion for eyes and neck/shoulder. Wilcoxon signed rank test was also used to compare high and low contrast of physical workload across all the workstations. Bonferroni corrections were performed for multiple comparisons ($n = 10$) for each percentile and muscular rest. A p-value of <0.005 (two-tailed) was then considered significant.

Table 1. Positive and negative reviews of the five different workstations by 29 non-presbyopic participants performing computer work ('visual search' and 'drag and align'). Also included are group means (SD in brackets) of perceived exertion for eyes and neck/shoulders at the different workstations, and participants' ratings of the best and worst workstations are presented.

			Positive assessments		Negative assessments	
Work station	Perceived exertion eyes	Perceived exertion neck/ shoulders	Ratings of best station ($N=24$)[a]	Positive reviews ($N=29$)	Ratings of worst station ($N=23$)[b]	Negative reviews ($N=29$)
27	3.5 (2.4)	2.2 (1.5)	9	• Felt natural, lighter, easiest, most comfortable. ($N=5$) • Did not have to twist so much. ($N=3$) • Saw better, did not have to look, smaller area, shorter distance. ($N=3$)	0	• Had to bend my neck more, more tired in the neck. ($N=2$) • Couldn't fit so much in. ($N=1$) • Annoying to sit still. ($N=1$)
27 L	3.7 (2.1)	4.5 (2.0)	1	• Easier to switch between two screens. ($N=1$) • Did not have to twist up and down as much. ($N=1$)	14	• Annoying to turn the neck. ($N=15$) • More annoying with two screens with a joint in the middle. ($N=5$)
27 V	3.5 (2.1)	3.2 (1.7)	0	• Sight was easier, saw better. ($N=1$) • Easier to switch between two screens, easier to scan off a small screen. ($N=1$) • Did not have to twist my head so much. ($N=1$)	5	• Annoying with two screens. ($N=3$) • Joint in the middle ($N=3$) • Never looked straight ahead, no main screen, had to move my gaze. ($N=3$) • Far out to the screen edges. ($N=1$) • Both screens were troublesome. ($N=1$)
49–70	3.7 (2.1)	3.1 (1.8)	2	• Did not have to twist that much. ($N=3$) • A bit easier. ($N=1$) • Favourite/best: the long screens. ($N=1$)	4	• Came too close. ($N=6$) • Had to twist my neck a lot. ($N=3$) • Difficult to see the edges and corners due to curved screen. ($N=3$) • Hard on the neck. ($N=1$)
49–90	3.4 (1.8)	2.1 (1.7)	12	• Better sight distance, better overview, saw better. ($N=10$) • Did not have to twist so much. ($N=3$)	0	• Difficult to see the edges/ corners with the curved screen. ($N=4$) • Distance too far. ($N=3$)

[a]Three participants rated more than one as the best workstation. Two participants did not express a best workstation.
[b]One participant rated more than one as the worst workstation. One person expressed that no one was the worst. Four participants did not express any worse workstation.

Table 2. Muscular load in the right trapezius shoulder and the right trapezius neck during computer work (visual search and drag and align) at five different workstations. Mean values and standard deviations (SD) are shown.

	Distribution (percentile)	27 $N=29$	27L $N=27$	27V $N=29$	49–70 $N=28$	49–90 $N=29$	Overall p-value (Friedman)
Muscular load							
Trapezius shoulder							
Activity (*% MVE*)	50th	0.8 (0.9)	1.0 (0.8)	0.9 (1.1)	0.7 (1.0)	0.6 (0.7)	<0.001
	90th	1.8 (1.4)	2.1 (1.6)	2.2 (1.8)	1.7 (1.5)	1.6 (1.2)	0.02
Rest (*% time*)		56 (30)	45 (32)	46 (22)	60 (29)	61 (26)	<0.001
Trapezius neck							
Activity (*% MVE*)	50th	3.1 (1.9)	3.3 (1.8)	3.3 (2.0)	3.1 (2.0)	2.9 (1.8)	0.001
	90th	4.7 (2.7)	5.1 (2.7)	5.2 (3.1)	4.8 (2.9)	4.4 (2.5)	<0.001
Rest (*% time*)		2.4 (6.0)	1.8 (5.3)	0.9 (2.1)	1.4 (2.8)	2.6 (6.2)	0.76

All pairwise comparisons (Wilcoxon) that were statistically significant are shown with a clamp. The p-values for all these comparisons were below 0.005 (ranging from <0.001 to 0.005; Bonferroni corrected).

Results

Muscular load between workstations

The results of muscular load recordings in the right trapezius shoulder and trapezius neck between workstations are presented in Table 2. Generally, the muscular loads in both trapezius shoulder and trapezius neck were lower at workstation 49–90 compared to the other four workstations. The group means of the 50th (median) percentile in both the trapezius shoulder and trapezius neck muscles were significantly higher at workstation 27L compared to 49–90 ($p < 0.001$). The same was true for the 90th (peak) percentile in the trapezius neck. The group mean of the peak percentile in the trapezius neck muscle was also significantly higher at workstation 27V compared to 49–90 ($p < 0.001$). The proportion of muscular rest in the trapezius shoulder muscle was very high at all workstations (45–61% of time). Still, the proportion of muscular rest at workstations 27L and 27V were lower than workstations 49–70 and 49–90 (p-values ranging from <0.001 to 0.004). The proportion of muscular rest in the trapezius neck muscle was substantially lower than in the trapezius shoulder muscle for all workstations (0.9–2.6% of time), but with no significant differences between the workstations.

Postures between workstations

Head and neck postures between workstations are presented in Table 3. The head was kept in the most upright position at workstation 49–90 compared to workstation 27, 27L and 27V (p-values ranging from <0.001 to 0.005; median values). The head was mostly flexed at workstation 27. The group mean of the median value of neck flexion was significantly lower at workstation 49–90 compared to workstations 27 ($p < 0.001$) and 27V ($p < 0.001$). The back flexion showed no significant differences between the workstations (data not shown).

Comparisons between high and low contrast

There was no significant difference concerning the muscular load in the trapezius shoulder during work with high and low contrast (Figure 6). However, the muscular load in the trapezius neck was significantly higher during low contrast compared to the high: the group means of the median load 3.5% of MVE (SD 2.3) vs. 3.1% of MVE (SD 1.8), the group means of the peak load 5.4% (SD 3.3) vs. 4.5% (SD 2.4); all p-values <0.001. The number of mouse clicks (all five workstations merged) were higher for high contrast compared to low: group means of 438 (range 218–760) vs. 239 (range 110–423).

Perceived exertions

The results of the perceived exertions for the eyes and neck/shoulder region are presented in Table 1. The group mean of eye exertion after working at each workstation varied between 3.4 and 3.7 on the Borg CR-10 scale. There were no significant differences between the workstations.

Table 3. Head, neck, and upper back postures during computer work tasks (visual search and drag and align) at five different workstations. Mean values and standard deviations (SD) are shown.

	Distribution (percentile)	27 N = 29	27L N = 27	27V N = 29	49–70 N = 28	49–90 N = 29	Overall p-value (Friedman)
Posture							
Head flexion (◦)							
	50th	8.7 (4.6)	7.8 (4.7)	7.9 (4.6)	7.5 (4.7)	6.5 (4.3)	<0.001
	90th	13 (5.2)	13 (5.5)	13 (5.2)	12 (5.2)	10 (4.7)	<0.001
Neck flexion (◦)	50th	5.9 (7.7)	4.7 (8.7)	6.0 (7.5)	4.6 (8.7)	3.1 (7.7)	<0.001
	90th	10 (7.4)	10 (8.3)	11 (7.3)	10 (9.2)	7.7 (7.7)	<0.001
Upper back (◦)	50th	1.4 (0.3)	1.4 (0.4)	1.5 (0.5)	1.4 (0.3)	1.4 (0.3)	0.97
	90th	4.4 (1.5)	4.9 (2.2)	4.9 (2.3)	4.7 (2.0)	4.4 (1.6)	0.11

All pairwise comparisons (Wilcoxon) that were statistically significant are shown with a clamp. The p-values for all these comparisons were below 0.005 (ranging from <0.001 to 0.005; Bonferroni corrected).

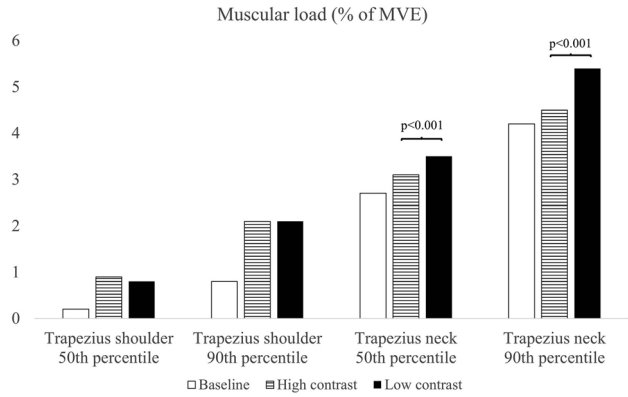


Figure 6. The 50th and 90th percentiles of muscular load (% of maximal voluntary electrical activity; MVE) during baseline, high contrast and low contrast for the right trapezius shoulder and trapezius neck across all five workstations.

The group mean of exertion in the neck/shoulder region after working at each workstation varied between 2.1 and 4.5. The exertion was significantly higher for workstation 27 L compared to all other workstations ($p < 0.001$ for all comparisons). Workstations 27 and 49–90 had the lowest group means.

Subjective ratings and comments

Twelve of the 24 participants rated 49–90 as the best workstation setup, while 9 of the 24 participants rated 27 as the best (three participants rated more than one setup as the best and two rated none as the best; Table 1). Fourteen of the 23 participants rated 27 L as the worst setup (one participant rated more than one setup as the worst workstation; one expressed that no one was the worst station, and four did not express any worst workstation).

Discussion

The results of the present study showed a lower muscular load in both the trapezius shoulder and the trapezius neck for workstation 49–90 compared to 27 L and 27 V, with statistically significant differences for trapezius shoulder and trapezius neck between workstations 49 and 90 and 27 L. The perceived exertions and the subjective ratings were in line with the results of the objective measurements. Thus, the perceived exertion in the neck and shoulder region was lower for workstations 49–90 and 27 compared to 27 L. Furthermore, the subjective ratings of best workstation were the highest for workstations 49–90 and 27.

Muscular load

The muscular load in the trapezius shoulder and trapezius neck showed low values, which means that the computer work performed in the present study was not physically demanding. However, with long-time exposure, such low-level sustained muscular load may cause WMSDs in the long-term (Arvidsson et al. 2021). In addition, there was a high proportion of muscular rest (<0.5% of MVE) in the trapezius shoulder, 45–61% of time across all workstations. A value below 5% of time has been reported to be a risk for the development of WMSDs (Amir et al. 2022). Thus, the recorded values of muscular rest in the trapezius shoulder in the present study implies a small risk for developing WMSDs. However, for the trapezius neck muscle, the low values of muscular rest indicate very small opportunities for the muscles to rest (values <3% of time across all workstations). One explanation may be that the trapezius neck muscles are constantly activated due to high eye strain without possibilities to achieve muscular rest. A further explanation may be that the trapezius neck muscle is active due to its function of being a postural muscle to hold the head in an upright posture.

Postures

Workstation 49–90 had the most upright head position compared to the other workstations. This could be explained by the difference in the viewing angle between 49 and 90 and the others as well as the distance to the screen being 20 cm longer than for the others. Workstation 27 showed a more flexed head position compared to 27 L/27V/49–70. We have no explanation for this. However, we speculate that it may depend on the need to rotate one's head when using multiple or large screens, and that it is necessary to raise the chin to be able to maintain the convergence during head rotation. This results in a more upright head position. This speculation is based on a previous study carried out by Asharlous et al. They concluded variations in convergence ability (i.e. reflex function)

during multiple gaze positions (Rubin 2018). The back flexion showed no significant differences between the workstations. This was expected as we instructed the participants to sit upright and as far as it was possible to keep the position throughout the experiment.

Comparisons between high and low contrast

Visual objects with a low contrast have a negative impact on the eyes' functional ability compared to visual objects with a high contrast (West et al. 2002; Estember et al. 2015). Since eyestrain has been demonstrated to be associated with muscle activity in the neck and shoulders (Zetterberg et al. 2017), it would be expected to see a higher muscular load in both the trapezius shoulder and trapezius neck during low contrast. However, in the present study, this could only be seen in the trapezius neck, with a higher muscular load during low contrast compared to high (Figure 6). No such difference for the trapezius shoulder could be seen. In fact, the muscular load was basically the same for both contrasts. We speculate that the muscular load in the trapezius shoulder during high contrast was mainly due to arm movements during the mouse clicking, with almost twice as many mouse clicks during high contrast compared to low. Furthermore, the proportion of muscular load in the trapezius shoulder during both high and low contrast compared to baseline was higher than the proportion of muscular load in the trapezius neck during contrasts compared to baseline (Figure 6). Thus, the results in the current study indicate that the relationship between the eyes and muscular load in the neck/shoulder region is more related to the trapezius neck than to the trapezius shoulder muscles.

Perceived exertions and subjective ratings

Overall, the participants in the current study favoured workstations 49–90 and 27 both in terms of perceived exertion in the neck/shoulder region and the highest subjective ratings. Regarding viewing distances, the participants preferred different distances due to their subjective comfort when using the large curved 49-inch screens. Half of them stated that it was better viewing distance for 49–90, while some of them thought it was too long. This indicates that there is a need for flexibility for best individual comfort when it comes to viewing distances. This is in line with a study performed by Jaschinski et al. who concluded that it is essential that individuals have the possibilities to choose their favourite viewing distance, to reduce visual strain (Jaschinski et al. 1999). However, having full flexibility for viewing distance when using large screens is reduced since today's standardised table depth is 80 cm. The participants in the present study also stated that they did not have to turn their head as much when using 49–90 compared to the other workstations (with a viewing distance of 70 cm). Thus, a shorter viewing distance may lead to neck issues due to larger head rotations (to be able to see far out laterally). For prevention, a recommendation would be to avoid using the outer edges of the screen when the large screen is placed at a shorter viewing distance.

27L had the highest score for perceived exertion in the neck/shoulder region compared to all other workstations. The most prominent comment was 'strenuous to rotate the neck'. The participants' experiences in the present study are in line with an Estember et al. study, where their participants reported higher twisting scores for head and neck at an L-shaped monitor setup compared to both single- and V-shaped setups, during different computer tasks (Chiu et al. 2002). In the same study, the participants reported the lowest twisting scores for head and neck for the single monitor setup compared to the dual monitor setup (L- and V-shaped). The results for the single monitor setup in the present study are the same as in Estember's study, where several participants stated that they preferred 27 compared to multiple screens (27 L and 27 V), with lower perceived exertion in the neck and shoulders compared to 27 L and 27 V. Concerning the ratings of 'best workstation', 9 participants rated 27 as the best one, while no participants rated 27 L as the best and only one participant rated 27 V as the best.

Methodological considerations

Method triangulation was used to interpret the data to obtain valid and robust results. The reliability and robustness of the results were assured by merging the results from the objective, technical recordings of the muscular load in the neck/shoulders and the head/neck postures with both perceived exertions in the neck/shoulders and the participants' subjective ratings and assessments of the different workstations. Additional strengths in the study were that an equal number of women and men participated in the study and that there was a wide distribution in both height and weight among the participants. Finally, to avoid the effects of fatigue, the order in which the five workstations were conducted were randomised.

In the present study, we used inclinometry to record head and back postures. A limitation with this method is that it is not possible to record protraction, i.e. an extended neck with a fixed back. Protraction has been

discussed in the literature as a risk factor for developing neck/shoulder complaints (47). Moreover, with our methodology choice, we could not measure rotation of the head and neck. Thus, the comments about neck rotation could not be verified by the technical recordings. However, we believe that the subjective comments are trustworthy. Several factors have been adjusted for in the present study. This is both a strength and a limitation at the same time. For example, the instruction from the research leaders was to sit in the same position with the body straight ahead throughout the experiment. With such a position, the participants were forced to turn their heads quite a lot to be able to see further out laterally, especially when working at the 49-inch screens. With this instruction, on the other hand, deviations in results due to the research participants sitting as they preferred during the experiment were eliminated.

The results in the present study are based on non-presbyopic participants. Consequently, our proposed recommendations are aimed at this group of people. Recommendations for presbyopic people will be proposed soon from the results in another laboratory study that has already commenced. The presbyopic participants are using three different types of progressive addition lenses (PALs) in the same work tasks as in the present study, at two different screen sizes.

Conclusions

Overall, the 49-inch curved screen with a viewing distance of 90 cm was both quantitatively and qualitatively rated as the best setup of all the five workstations included in the study. The 49-inch screens were also objectively and subjectively rated to be better than 2×27-inch screens. The 27V was subjectively favoured compared to 27L.

Based on the results from the present study, our proposed recommendations on size, viewing distance and screen setup for non-presbyopic humans, to reduce the risk of visual and physical ergonomic complaints during computer work are:

- Use one curved 49-inch screen with a viewing distance of 90 cm.
- If the table is not deep enough for a viewing distance of 90 cm for the 49-inch screen or if the person prefers a shorter distance due to personal comfort, a somewhat shorter viewing distance may be used. In those cases, do not use the screen surface far out laterally.
- Use a curved 49-inch screen rather than 2×27-inch screens. If a 2×27-inch screen setup is the only alternative, choose 27V, not 27L.

The results from the present study apply to non-presbyopic participants. These results are based on technical recordings of muscular load in the trapezius shoulder and trapezius neck as well as subjective exertions in shoulder/neck and subjective ratings of the different screen setups.

Note

1. The eye is looking straight ahead with the visual axis parallel to the sagittal plane of the head.

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Data availability statement

The data that support the findings of this study are available from the corresponding author on reasonable request.

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